The irrationality exponent of $\log(3/2) + \sqrt{2}$ is at most four

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October 12, 2026

Abstract

We prove that $\log(3/2) + \sqrt{2}$ has irrationality exponent at most four. More generally, we remove the algebraic-integrality hypothesis from our previous affine theorem while retaining its other hypotheses. Additional column factors clear the fixed algebraic denominators in the preceding determinant construction. Together with the original row and column factors, they give an algebraic-integer determinant whose norm supplies the arithmetic lower bound. Their normalized logarithmic cost can be made arbitrarily small through the parameter choices, so the interpolation and analytic estimates from our previous proof still give the bound four. A Lean formalization accompanies the proof.

1 Introduction and main result

The number $\log(3/2) + \sqrt{2}$ is transcendental by the Hermite–Lindemann theorem [4]: $\log(3/2)$ is transcendental, and adding the algebraic number $\sqrt{2}$ preserves transcendence. We study the quality of its rational approximations and prove that its irrationality exponent is at most four.

For an irrational real number x, its irrationality exponent $\mu(x)$ is the supremum of the positive real ν for which $0 < |x - p/q| < q^{-\nu}$ holds for infinitely many reduced fractions p/q with $q > 0$. We prove the following bound.

Theorem 1.1. *For every real $\nu > 4$, there is an integer $Q \geq 2$ such that*

$$q^{-\nu} \leq \left| \log(3/2) + \sqrt{2} - \frac{p}{q} \right| \qquad (p \in \mathbb{Z},\ q \in \mathbb{N},\ q \geq Q). \tag{1}$$

Consequently, $\mu(\log(3/2) + \sqrt{2}) \leq 4$.

The threshold depends on ν and is uniform in p, q. Fractions in (1) need not be reduced.

Our previous affine theorem [1] applies to algebraic-integer data in a degree-four field with two compatible embeddings. The base $3/2$ introduces a denominator into the interpolation matrix. Clearing it costs a factor that grows with the matrix height, so we must account for that factor when choosing the earlier parameters.

We remove the algebraic-integrality hypothesis from our previous theorem while retaining its other hypotheses, including the two compatible embeddings and the slope bound. Additional column factors clear the fixed algebraic denominators alongside the row and column factors of our previous construction. Their normalized logarithmic cost can be made arbitrarily small by choosing the dimension and approximation weights. We then use the interpolation and analytic estimates from our previous proof [1], which builds on OpenAI's determinant method [2] and our

logarithmic extensions [5, 6]. Sections 2–4 give the general argument, followed by the specialization to $\log(3/2) + \sqrt{2}$.

Marcovecchio's nonquadraticity table [3] gives $\mu_2(\log(3/2)) < 9.460812$. With that paper's minimal-polynomial height, this implies the rational-exponent bound 18.921624 for $\log(3/2) + \sqrt{2}$; Section 5.1 gives the conversion. Theorem 1.1 improves this bound to four.

2 The affine theorem with algebraic denominators

Write $\mathrm{Emb}(F) = \mathrm{Hom}_{\mathbb{Q}\text{-alg}}(F, \mathbb{C})$. The ring of integers is denoted by $\mathcal{O}_F$.

Theorem 2.1. *Let $[F : \mathbb{Q}] = 4$ and $a, u, v \in F$. Suppose an embedding σ_0 satisfies $\sigma_0(a) = b_0 \in \mathbb{R}$, with $b_0 > 0$ and $b_0 \neq 1$. Let $x \in \mathbb{R} \setminus \{0\}$. Suppose a selected set $G \subset \mathrm{Emb}(F)$ has cardinality two and*

$$\exp(\sigma(u)(x - \sigma(v))) = \sigma(a), \qquad |\sigma(u)| \leq 2 \quad (\sigma \in G). \tag{2}$$

Then, for every real $\nu > 4$, there is $Q \in \mathbb{N}$, $Q \geq 2$, such that $q^{-\nu} \leq |x - p/q|$ for every $p \in \mathbb{Z}$ and $q \in \mathbb{N}$, $q \geq Q$. In particular, x is irrational and $\mu(x) \leq 4$.

Other embeddings may also satisfy (2), and σ_0 need not belong to G. Embeddings are counted as distinct maps even when their values on a, u, v coincide. The result does not require the remaining two embeddings to be real.

Choose positive integers d_a, d_n such that

$$d_a a, \quad d_n u, \quad d_n uv \ \in \mathcal{O}_F. \tag{3}$$

Every element of a number field has a positive integer multiple in its ring of integers. Taking a product of denominators clears the last two elements simultaneously. These choices depend only on the fixed algebraic data, not on any approximating fractions or matrix height.

2.1 Interpolation and determinant estimates

We recall the interpolation and determinant estimates used to prove our previous affine theorem [1]. For a fixed $\nu > 4$, suppose there are approximations $|x - p/q| < q^{-\nu}$ with arbitrarily large q. Choose finitely many p_i/q_i with $q_i \geq 2$ and $p_i \neq 0$, and put $w_i = \lceil \log q_i \rceil$, $w_* = \min_i w_i > 0$. Nonzero numerators are available by increasing the initial threshold, since $x \neq 0$. There are rational shape parameters

$$0 < \theta < A < B < 1, \quad C > 1, \quad B < C^{-1}, \quad B < C\theta < 1, \tag{4}$$

and $0 < \eta < 1$, for which

$$g = \nu(A(1 - \eta) - \theta) - 2(1 - \theta) > 0. \tag{5}$$

Choose $F_0 > 2/\theta$, a dimension $m \geq 1$, and set

$$K = \lfloor C^m \rfloor, \qquad w_0 = B^{-m}, \qquad v_0 = 2K\theta^m w_0. \tag{6}$$

The finitely many weights w_i are then chosen successively large enough to satisfy the weight-separation hypotheses of weighted logarithmic interpolation in our previous paper [1, Sections 4–5]. These hypotheses compare finitely many products of earlier weights; an arbitrarily large next denominator suffices. The strict volume ratio is $Kw_0\theta^m/v_0 = 1/2$.

For $H > 0$, use columns $(h, \gamma) \in \mathbb{N} \times \mathbb{N}^m$ with $w_0 h + \sum_{i=1}^m w_i \gamma_i \le H$, and rows (j, s, β) with $0 \le j < K$ and $v_0 s + \sum_{i=1}^m w_i \beta_i / \theta < H$. The number of rows is $M = M(H) > 0$. Define

$$T_i = \lceil F_0 w_i / v_0 \rceil, \qquad L_i(t) = \sum_{1 \le k < T_i} \frac{(-1)^{k+1}}{k} t^k, \tag{7}$$
$$r_{\sigma i} = \sigma(u)(p_i/q_i - \sigma(v)), \qquad z_\sigma = \sigma(a).$$

The matrix is

$$\mathcal{M}_\sigma(H)_{(j,s,\beta),(h,\gamma)} = z_\sigma^{jh} [t^s Z^\beta](1+t)^h \prod_{i=1}^m (Z_i + j r_{\sigma i} + L_i(t))^{\gamma_i}. \tag{8}$$

Weighted interpolation gives full row rank at σ_0 for a cofinal set of heights. Choose one nonzero full-row minor there and use its *same columns* at every embedding; its determinant is Δ_σ. Write

$$b_H = \frac{1}{MH} \sum_{\text{rows } r} \sum_i w_i \beta_{ri}, \qquad 0 \le b_H \le \theta.$$

Fix bounds $\kappa, R \ge 0$ and $\rho \ge 100$ such that, at all embeddings,

$$\log \max(1, |\sigma(a)|) \le \kappa,$$
$$|\sigma(u)|(|x| + 1 + |\sigma(v)|) \le R,$$
$$100|x| \le \rho, \qquad 100|\sigma(u)(x - \sigma(v))| \le \rho.$$

Finite sums or maxima provide such bounds before the approximations are selected. Put $\Lambda = \log 4 + 4$ and $c = \log 2/4$, and define

$$E_0 = \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_i \frac{1}{w_i} + \frac{\theta}{w_*}, \tag{9}$$
$$U(S) = \frac{\log 2}{v_0} + \frac{\kappa K + \log(3/2)}{w_0} + \frac{\log(2(KS+2))}{w_*},$$
$$V = \frac{\nu}{F_0} + \frac{2\log 2}{v_0} + \frac{\rho K}{w_0} + \frac{\log 4 + \log(2K) + \nu + \log(200K) + \log(\rho/100)}{w_*}.$$

Our previous estimates [1, Sections 7–8] give remainders $r_a(H), r_c(H) \to 0$ and a collision rate $C_H \to C_\infty$, where

$$C_\infty = \frac{c\eta^2 K \theta^m}{(m+1) v_0 A^m} = \frac{c\eta^2}{2(m+1)} (B/A)^m, \tag{10}$$

such that every nonzero selected minor satisfies

$$\frac{\log |\Delta_\sigma|}{MH} \le V + r_a(H) + \frac{\log 2}{w_*} + \max\{-C_H, -\nu(A(1-\eta) - b_H)\}, \qquad \sigma \in G, \tag{11}$$
$$\frac{\log |\Delta_\sigma|}{MH} \le U(R) + r_c(H), \qquad \sigma \notin G. \tag{12}$$

These estimates apply to the matrix (8) before denominator clearing. Their hypotheses concern the complex bases and centres, the exponential identities, the approximation errors and the numerical parameters. Interpolation likewise permits arbitrary additive centres. Both results therefore apply to the present algebraic data.

3 Clearing denominators

Let $D_i = \operatorname{lcm}(1, \dots, T_i)$ and $e_i = \lfloor H/w_i \rfloor$. The original row and column factors are

$$\prod_i D_i^{e_i} q_i^{-\beta_i} \quad \text{and} \quad \prod_i q_i^{\gamma_i}.$$

Their product over the selected square matrix is a positive rational number $\mathscr{C}_H$. The elementary clearing-cost calculation, which is an inequality about these numerical factors, gives

$$\log \mathscr{C}_H \le MH(1 - b_H + E_0). \tag{13}$$

We next clear the denominators of the fixed algebraic data.

Add to each selected column (h, γ) the integer factor

$$c(h, \gamma) = d_a^{Kh} d_n^{\sum_i \gamma_i}, \tag{14}$$

and let A_H be the product of these extra factors over the selected columns. The exponent Kh bounds jh for every row index $j < K$.

Lemma 3.1. *There is a single nonzero algebraic integer $Z_H \in \mathcal{O}_F$ such that*

$$\sigma(Z_H) = \mathscr{C}_H A_H \Delta_\sigma \qquad (\sigma \in \operatorname{Emb}(F)). \tag{15}$$

In particular, all four selected determinants are nonzero.

Proof. Put $n_i = u(p_i - q_i v)$ and $N_i = d_n n_i$. Equation (3) makes $N_i = (d_n u)p_i - (d_n uv)q_i$ integral. If some $\beta_i > \gamma_i$, the matrix entry is zero. Otherwise put $\delta_i = \gamma_i - \beta_i$. After extracting Z^β and applying the original factors, the entry is the embedded image of

$$a^{jh} \prod_i \binom{\gamma_i}{\beta_i} [t^s](1+t)^h \prod_i D_i^{e_i} (jn_i + q_i L_i(t))^{\delta_i}.$$

The additional factor (14) clears the remaining denominators because

$$\begin{aligned} d_a^{Kh} a^{jh} &= d_a^{(K-j)h} (d_a a)^{jh}, \\ d_n^{\gamma_i} (jn_i + q_i L_i)^{\delta_i} &= d_n^{\beta_i} (jN_i + d_n q_i L_i)^{\delta_i}. \end{aligned}$$

All exponents are nonnegative. The polynomial $D_i(jN_i + d_n q_i L_i)$ is integral coefficientwise, and $\delta_i \le \gamma_i \le e_i$, so its δ_ith power is cleared by $D_i^{e_i}$. Products, binomial coefficients and coefficient extraction preserve integrality.

Thus these entries define one square matrix over $\mathcal{O}_F$. Its determinant is Z_H, and determinants commute with every embedding. This proves (15). At σ_0 its image is nonzero, since both factors are positive and the selected minor is nonzero. Hence $Z_H \neq 0$, and injectivity of every field embedding proves the last assertion. Each embedding uses the columns chosen at σ_0. □

Lemma 3.2. *With*

$$\tau = \frac{K \log d_a}{w_0} + \frac{\log d_n}{w_*}, \tag{16}$$

one has $0 \le \log A_H \le MH\tau$.

Proof. Every column satisfies $h \le H/w_0$ and $\sum_i \gamma_i \le H/w_*$. Apply these inequalities to the logarithm of (14), then sum over M columns. The logarithms of the positive integer denominators are nonnegative. The column budget bounds the sum of all transverse exponents, giving the single term $\log d_n/w_*$. □

The nonzero integer norm of Z_H gives

$$0 \le 4\log(\mathscr{C}_H A_H) + \sum_\sigma \log|\Delta_\sigma|.$$

Using (13), Lemma 3.2, and the estimates at two good and two remaining embeddings, then dividing by $2MH > 0$, gives

$$-2(1-b_H) - 2(E_0+\tau) - U(R) - r_c(H) \le V + r_a(H) + \frac{\log 2}{w_*} + \max\{-C_H, -\nu(A(1-\eta)-b_H)\}. \tag{17}$$

Thus denominator clearing adds at most 2τ to the normalized arithmetic cost. The analytic estimates apply to the Δ_σ.

4 Absorption and the contradiction

Set

$$E = E_0 + \frac{K\log d_a}{w_0}, \qquad \widehat{R} = 2d_n^2(R+2).$$

For $K \ge 1$, $R \ge 0$, and $d_n \ge 1$,

$$2(K\widehat{R}+2) \ge 2d_n^2\, 2(KR+2).$$

Indeed the difference is $8d_n^2(K-1)+4 \ge 0$. Taking logarithms and dividing by $w_* > 0$ proves

$$U(R) + \frac{\log 2 + 2\log d_n}{w_*} \le U(\widehat{R}). \tag{18}$$

This enlargement pays the transverse denominator cost and the good-embedding multiplier simultaneously.

We now justify the required parameter choices with these fixed costs. First choose shape parameters with (5); for example take rational $t < 1$ close enough to one that $\nu t > 2(1+t)$, put $\theta = t^2$, choose rational $\theta < A < B < t$ close to t, then choose $C \in (B/\theta, 1/B)$ and a sufficiently small $\eta > 0$. Choose $0 < \epsilon < g$ with $\epsilon \le 1/2$, then choose $F_0 > 2/\theta$ with $\nu/F_0 < \epsilon/3$.

Choose m sufficiently large that

$$\frac{2\Lambda F_0 m + 3\log 2}{v_0} + \frac{(\rho+\kappa+2\log d_a)K + \log(3/2)}{w_0} < \epsilon/3, \qquad C_\infty > 3. \tag{19}$$

This is possible since $K/w_0 \le (CB)^m \to 0$, $m/v_0 \to 0$ because $C\theta/B > 1$, and $C_\infty \to \infty$ by (10). Next choose the geometric slack required for interpolation and increase the initial approximation-weight threshold so that all remaining inverse-weight terms in $2E + U(\widehat{R}) + V$ sum to less than $\epsilon/3$. Their coefficients are fixed at this point, and $\sum_i 1/w_i \le m/w_*$. Later denominators can still be chosen successively large enough for separation. Consequently

$$2E + U(\widehat{R}) + V < g, \qquad 2 + 2E + U(\widehat{R}) + V < C_\infty. \tag{20}$$

We have now fixed the dimension, weights and finite approximation tuple. The terms $K \log d_a/w_0$ and $\log d_n/w_*$ are small by these choices and remain constant as H tends to infinity.

By (18), (17) implies

$$-2(1-b_H) \le S_H + \max\{-C_H, -\nu(A(1-\eta)-b_H)\},$$

where $S_H = 2E + U(\widehat{R}) + V + r_a(H) + r_c(H)$. For all sufficiently large H, (20) yields $S_H < g$ and $2 + S_H < C_H$. If the maximum is the approximation term, rearranging gives

$$g + (\nu-2)(\theta - b_H) = \nu(A(1-\eta)-b_H) - 2(1-b_H) \le S_H < g,$$

a contradiction. If it is the collision term, it gives $C_H \le 2(1-b_H) + S_H \le 2 + S_H$, again a contradiction. Either argument works if the two terms are equal.

Interpolation supplies a nonzero minor at a cofinal set of heights, so this contradiction rules out arbitrarily large denominators with $|x - p/q| < q^{-\nu}$. This proves the bound in Theorem 2.1. A rational x would have exact representations with arbitrarily large unreduced denominators, contrary to that bound. Thus x is irrational. For each $\nu > 4$, only finitely many reduced fractions can violate it: denominators are bounded, and the inequality bounds their numerators as well. Hence $\mu(x) \le 4$.

5 Specialization and published comparison

Take $F = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, which has degree four, and put

$$a = 3/2, \qquad u = 1, \qquad v = \sqrt{2}, \qquad x = \log(3/2) + \sqrt{2} > 0.$$

The two embeddings fixing $\sqrt{2}$ and sending $\sqrt{3}$ to its two signs are distinct. They satisfy (2), since $\exp(x - \sqrt{2}) = 3/2$, and their slopes have absolute value one. The positive distinguished base is $3/2 \ne 1$. We may choose $d_a = 2$, $d_n = 1$ in (3). Theorem 2.1 proves Theorem 1.1.

5.1 Height conversion

For coprime p, q, $q > 0$, the quadratic number $U = p/q - \sqrt{2}$ has primitive irreducible integer polynomial

$$q^2X^2 - 2pqX + (p^2 - 2q^2).$$

Primitivity follows from $\gcd(q^2, p^2 - 2q^2) = 1$; irreducibility follows from $\sqrt{2} \notin \mathbb{Q}$. Its coefficient height obeys $H(U) \ge q^2$. For approximations with $|x - p/q| < 1$, it also obeys $H(U) \le C_x q^2$, where $C_x > 0$ is fixed. Marcovecchio's definition and table [3, pp. 147, 151, 182] imply, for every $s > 9.460812$ and all sufficiently large such heights,

$$|x - p/q| = |\log(3/2) - U| > H(U)^{-s} \ge C_x^{-s} q^{-2s}.$$

Choosing $9.460812 < s < \nu/2$ and absorbing the fixed constant gives the comparison bound $\mu(x) \le 18.921624$. The factor two comes from the quadratic growth of coefficient height.

6 Further questions

Dirichlet's theorem and Theorem 1.1 give $2 \le \mu(\log(3/2) + \sqrt{2}) \le 4$. Can the upper bound be reduced to two? Our argument leaves this question unresolved. It gives an eventual threshold for each $\nu > 4$; obtaining an explicit threshold or a lower bound at the critical exponent four would require further estimates.

A Formalization

The accompanying Lean formalization proves Theorem 2.1 and its specialization in Theorem 1.1. The arithmetic construction gives one matrix over $\mathcal{O}_F$ and proves that its determinant maps to the positive scalar multiple in (15) at every embedding. The parameter construction uses $\log d_a$ for the base-denominator cost and $\widehat{R}$ for the coarse bound.

The formalization derives irrationality and the standard `LiouvilleWith` consequence from the approximation bound: $\neg\texttt{LiouvilleWith}(\nu, x)$ for every $\nu > 4$. We absorb the constant in that definition by applying the approximation bound at $\nu' = (4+\nu)/2$ and using the smaller-exponent approximations supplied by Mathlib.

The general affine bound and its specialization were compared with independently written specifications. For $\log(3/2)+\sqrt{2}$, the comparison covers the eventual and explicit-threshold bounds, irrationality, the irrationality exponent, and the `LiouvilleWith` consequence.

Comparator's two kernel checks and a separate stock Lean 4.34.1 replay accepted the proofs. The export uses only `propext`, `Classical.choice`, and `Quot.sound` as axioms. The companion `VERIFY.md` records the statement correspondence, source and tool versions, reproduction instructions, and the scope and limitations of these checks.

Acknowledgements

OpenAI Codex agents assisted with mathematical development, formalization, manuscript preparation, and review.

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